MAHSA VATANPARAST, MD

Physician-Researcher • Email • Google Scholar • LinkedIn



Dr. Mahsa Vatanparast is a physician and medical AI researcher focused on uncertainty quantification for medical imaging AI. Her work applies conformal prediction and related methods to radiology report generation, glioma segmentation, molecular marker classification, and image registration, so that deep learning models can flag low-confidence predictions and become more trustworthy in clinical settings. Her work has been recognized with the SIIM Helen and Paul Chang Foundation New Investigator Award and the ASFNR John L. Ulmer Award.

Education

- **National Organization for Development of Exceptional Talents (NODET/Sampad)**, Iran's selective national program for gifted students (entry via competitive nationwide exam) (2013–2019)
- **Mashhad University of Medical Sciences**, Mashhad, Iran, *Doctor of Medicine (MD)* (2026)

Honors and Awards

- **Winner of the SIIM Helen and Paul Chang Foundation New Investigator Award (US)** for CONRep: Uncertainty-Aware Vision-Language Report Drafting Using Conformal Prediction and **winning a $1,250 prize** (2026)
- **Winner of the John L. Ulmer Award at ASFNR (US, CA)** for CONSeg: Voxelwise Uncertainty Quantification for Glioma Segmentation Using Conformal Prediction and **winning a $1,000 prize** (2025)
- **Rank 3** among presented abstract in Sharif AIH Congress and **won a $500 prize** (2025)
- **Ranked 1st** among the best student ideas in the final round of the "Artificial Intelligence in Medical Sciences Idea Challenge" **and won a $400 prize** (2024)
- **Ranked 1st** among student presentations at the Annual Congress of the Iranian Medical Council (2024)

Research Experience

• Research Fellow, Transplant Research Center, Montaserieh Hospital

- Co-led **10+ published research projects** in **AI for medical imaging** since 2024
- Serve as a **peer reviewer** for BioMedical Engineering Online (Q1), European Journal of Medical Research (Q2), and Karbala International Journal of Modern Science (Q2)
- Co-founded and led NeuroAIHub, an open-source **AI-driven registry for neuroradiology dataset discovery** (GitHub, Demo, web app, published in the American Journal of Neuroradiology)

Grants

• Research grants: **15+ research grants** funded by the **Vice Chancellery for Research, Mashhad University of Medical Sciences (MUMS)**.

• Recipient of a **$2,000 grant** awarded by the **Transplant Research Center, Montaserieh Hospital**.

Publications

1. Elyassirad D, Gheiji B, **Vatanparast M,** et al. CONRep: uncertainty-aware vision-language report drafting using conformal prediction. J Imaging Inform Med. 2026.

2. Elyassirad D, Gheiji B, **Vatanparast M,** et al. Predicting IDH mutation in glioma patients using learning algorithms with conformal prediction. J Imaging Inform Med. 2026.

3. Gheiji B, Elyassirad D, **Vatanparast M,** et al. CONReg: uncertainty-aware medical image registration using conformal prediction. J Imaging Inform Med. 2026.

4. Elyassirad D, Gheiji B, **Vatanparast M,** et al. CONSeg: voxelwise uncertainty quantification for glioma segmentation using conformal prediction. AJNR Am J Neuroradiol. 2025.

5. Elyassirad D, Gheiji B, **Vatanparast M,** et al. Comparative analysis of 2D and 3D ResNet architectures for IDH and MGMT mutation detection in glioma patients. arXiv. 2024.

6. Hasanpour M, Elyassirad D, Gheiji B, **Vatanparast M,** et al. Predicting epidural hematoma expansion in traumatic brain injury: a machine learning approach. Neuroradiol J. 2025.

7. Rostami A, Elyassirad D, **Vatanparast M,** et al. Functional outcome and mortality predictors in patients with cerebral ischemic infarction after decompressive craniectomy: cross-sectional study. World Neurosurg. 2024.

8. Ahmadzadeh AM, Jafarnezhad A, Elyassirad D, **Vatanparast M,** et al. Improving radiomics-based isocitrate dehydrogenase 1 prediction in glioma patients using semi-supervised machine learning models. BMC Med Imaging. 2025.

9. Gheiji B, Moradi S, **Vatanparast M,** et al. NeuroAIHub: an AI-driven framework for automated curation and discovery of neuroradiology datasets. AJNR Am J Neuroradiol. 2026.

10. Ahmadzadeh AM, Ashoobi MA, Lomer NB, **Vatanparast M,** et al. Radiomics and deep learning models for predicting glioma p53 status: a diagnostic accuracy systematic review and meta-analysis. Clin Imaging. 2026.

11. Ahmadzadeh AM, Lomer NB, Ashoobi MA, Elyassirad D, Gheiji B, **Vatanparast M,** et al. Predicting 1p/19q codeletion status in glioma using MRI-derived radiomics: a systematic review and meta-analysis of diagnostic accuracy. AJNR Am J Neuroradiol. 2025.

12. Ahmadzadeh AM, Broomand Lomer N, Ashoobi MA, Elyassirad D, Gheiji B, **Vatanparast M,** et al. MRI-derived deep learning models for predicting 1p/19q codeletion status in glioma patients: a systematic review and meta-analysis of diagnostic test accuracy studies. Neuroradiology. 2025.

13. Ahmadzadeh AM, Ashoobi MA, Broomand Lomer N, Elyassirad D, Gheiji B, **Vatanparast M,** et al. Application of deep learning for predicting hematoma expansion in intracerebral hemorrhage using computed tomography scans: a systematic review and meta-analysis. Radiol Med. 2025.

14. Ghorbani M, Keykhosravi E, **Vatanparast M,** et al. Flow diverting stent monotherapy as the best choice in the treatment of intracranial blood blister-like aneurysms: a systematic review. Neurosurg Rev. 2024.

15. Ghorbani M, Keykhosravi E, **Vatanparast M,** et al. Traumatic middle meningeal artery aneurysm: a case report. Neurochirurgie. 2024.

Manuscripts Under Review

1. **Vatanparast M,** et al. Comparison of the performance of two decision-making support tools, ChatGPT and UpToDate, in the management of complex medical cases.

2. **Vatanparast M,** et al. Conformalized semi-supervised learning for medical imaging.

Congress

- **SIIM 2026**, Oral presentation, Winner of the Helen and Paul Chang Foundation New Investigator Award, *CONRep: Improving Reliability of Vision-Language Report Generation via Conformal Prediction*

- **SIIM 2026**, Poster presentation, *Uncertainty Quantification for Vision-Language Model Classification in Medical Imaging*
- **RSNA 2025**, Oral presentation, *NEUROAIHUB: An AI-Driven Framework for Automated Discovery and Retrieval of Neuroradiology Datasets*
- **ASFNR 2025**, Oral presentation, Winner of John L. Ulmer Award, *CONSeg: Voxelwise Uncertainty Quantification for Glioma Segmentation Using Conformal Prediction*
- **AIH Congress 2024**, Oral presentation, *Improving IDH Mutation Prediction using Conformal Prediction*
- **CARDIOHTMEDIC 2024**, Oral presentation, *Interpretable Deep Learning model for uncertainty-based left ventricle ejection fraction estimation in echocardiography*
- **ASFNR 2024**, Poster presentation, *Machine Learning and Deep Learning Performance in Predicting IDH Mutation, 1p/19q Co-Deletion, and MGMT Promoter Status Among Glioma Patients Using MRI*
- **Annual Congress of the Iranian Medical Council 2024**, Oral presentation, *Predicting Epidural Hematoma Expansion from Brain CT*

Technical Skills

- **Programming:** Python
- **Libraries & Frameworks:** PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy
- **ML/AI Methods:** Deep Learning (CNNs), LLM, VLM, ML, Conformal Prediction, Uncertainty Quantification
- **Medical Imaging:** MRI/CT Image Processing, Segmentation, Feature Extraction, Registration
- **Research:** Systematic Reviews, Network Meta-Analysis, Statistical Modeling
- **Content Creation:** Medical and Scientific Content Creation, Visual Content Design, Video Editing

Languages

- **Persian** (mother tongue)
- **English**: fluent
- **French**: fluent

References

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